Asmita Pal

Address: 1415 Engineering Drive, Madison, WI 53706
Website: asmita-pal.github.io | Linkedin : linkedin.com/in/asmita-pal/
Email: asmita.pal@wisc.edu | Phone: 480-516-3041

Summary

PhD candidate in Computer Architecture specializing in approximate computing, privacy-preserving traces, fully homomorphic encryption and interconnect security. Currently seeking research or engineering roles in performance modeling, memory subsystems, hardware-software co-design, and security.

Education

PhD in Computer Engineering — August 2018 - August 2025
University of Wisconsin - Madison

MS in Computer Engineering — August 2016 - August 2018
Utah State University

B.Tech. in Electronics and Instrumentation Engineering — July 2010 - June 2014
West Bengal University of Technology, India

Research and Work Experience

STACS Lab, University of Wisconsin - Madison — Madison, WI
Graduate Research Assistant — August 2018 – Present
Advisor : Joshua San Miguel

Side-channel attacks for mesh-based Network-on-Chips(NOC)

- Identified contention for Network-on-Chips for mesh interconnects using traffic patterns on Garnet in gem5.
- Developed temporal partitioning techniques inspired by SurfNoC to mitigate interconnect-level information leakage in SoCs.

Fully Homomorphic Encryption

- Developing analytical models for performance prediction of FHE-based applications. Followed by guided parameterization for ML-based predictors.
- Analyzed side-channels in applications encrypted using FHE (Fully Homomorphic Encryption), on the IBM HELayers framework.
- Identified spatial and temporal memory access patterns in encrypted queries that leak sensitive information.
- Demonstrated 90% accuracy in distinguishing database entries using ML classifiers, revealing a critical privacy vulnerability in secure computation.

Camouflage : Privacy-preserving Traces

- Addressed challenges in sharing confidential workloads for architectural research by balancing trace utility and privacy.
- Characterized utility of a trace as a function of program semantics that influence program behavior, as captured in memory traces.
- To ensure input privacy, developed a threat model based on input indistinguishability and show that the average security loss is 7.8% for obfuscated traces.
- Tested camouflaged encryption traces against secret key recovery.
- Proposed a trace-based obfuscation mechanism for workloads from SPEC2017, GAPBS and PERFECT Benchmark suites. Validated these traces against prefetchers and replacement policies on ChampSim.

As-Is Approximate Computing

- Applied reduced precision, random input sampling and tree-based sampling techniques for an approximate compute model, which is interruptive.
- Simulated such approximate applications (PERFECT and GAPBS Benchmarks) on a speculation-based architectural simulator for studying accuracy-runtime trade-offs.

Prefetching using Large Language Models

- Applied NLP-inspired techniques to memory access traces for learning temporal and spatial locality and predict addresses to prefetch.
- Compared existing hardware prefetchers vs LLM-predicted addresses for inference overhead and gap between coverage and accuracy.

Course Projects

Digit Recognition on FPGA using Multilayer Perceptron

- Designed processor, memory and accelerator modules to enable storage of a pre-trained model and offload computations to accelerator.
- Captured image and predicted image digit on the 7 segment display.

Precise encoding in Stochastic Computing for Neural Networks

- Proposed a new encoding scheme for Stochastic Numbers and implemented this new encoding format in MAC module of a small-scale Neural Network.
- Analyzed the energy-accuracy trade-off for LeNet300-100 and observed up to 5X energy improvement.

AMD Research — Boston, United States

Research Intern — June 2022 – Sept 2022

- Characterized last-level cache (LLC) traffic patterns for AI and HPC workloads on multicore CPU SoCs.
- Investigated heuristics and machine learning-based methods for improved cache management.
- Explored reuse-distance metrics to guide LLC replacement policies.
- Pending US Patent for reuse distance-based estimations for cache management.

AMD Research — Boston, United States

Research Intern — September 2020 – December 2020

- Collected memory and last level cache statistics using hardware performance counters.
- Used metrics based on software-hardware interaction and analyzed different kernels present in BERT to determine potential areas of acceleration.
- Generated kernel-specific traces to study performance and sensitivity to memory latency/bandwidth of this application on simulator.

Bridge Lab, Utah State University — Utah, United States

Graduate Research Assistant — August 2016 – July 2018

Design of a Process-Variation Aware Low Power GPU *Supervisor* : Koushik Chakraborty, Sanghamitra Roy

- Explored the role of process variation on register files for a GPU operating at Near Threshold Voltages.
- Analyzed the performance variation in GPGPU applications from AMD APP SDK suite, with varying register latency.
- Proposed a low overhead technique to overcome the effects of process variation on register latency. Exhibited $\sim$36% reduction in GPU energy consumption with marginal area and power overheads.

Course Projects

Speaker Diarization using Deep Neural Networks

- Extracted features from a conversation and used Deep Neural Networks to separate an audio signal according to speaker identity.
- Compared accuracy of the speaker diarization framework, using different deep neural networks (DNN) and convolutional neural networks (CNN).

Critical Thread Predictor

- Studied the statistical correlation of thread criticality metrics based on cache-misses and synchronization bottlenecks.
- Analyzed performance numbers for Splash2 Benchmarks on multi-core Sniper simulator.

Infosys Technologies Limited — Pune, India

System Engineer — July 2014 – August 2016

- Developed an automation framework for hardware verification, streamlining debugging processes and reducing manual verification time, to enhance team productivity.
- Managed client deadlines and service delivery, coordinating with cross-functional teams.

Publications

- **Asmita Pal**, Keerthana Desai, Rahul Chatterjee, Joshua San Miguel: **Camouflage: Utility-Aware Obfuscation for Accurate Simulation of Sensitive Program Traces**, published in ACM Trans. Archit. Code Optim. 21, 2, Article 36 (June 2024), 23 pages.
- Mitali Soni, **Asmita Pal**, Joshua San Miguel: **As-Is Approximate Computing**, published in ACM Trans. Archit. Code Optim. 20, 1, Article 3 (March 2023), 26 pages.
- **Asmita Pal**, Karthik Swaminathan, Subhankar Pal, Joshua San Miguel: **Characterizing Memory side channels in FHE applications** presented in DISCC Workshop *(held in conjunction with MICRO'22)*

- Pramesh Pandey, **Asmita Pal**, Koushik Chakraborty, Sanghamitra Roy: **Reliability and Uniformity Enhancement in 8T-SRAM based PUFs operating at NTC**, published in IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED) 2018.

- **Asmita Pal**, Aatreyi Bal, Koushik Chakraborty, Sanghamitra Roy: **Split Latency Allocator: Process Variation-Aware Register Access Latency Boost in a Near-Threshold Graphics Processing Unit**, published in J. Low Power Electronics 13(3): 419-427 (2017).

Awards

- Winner of N+1 Institute Google Reverse Pitch Competition - (AI Sustainability) at UW-Madison.
- Sarah and David Epstein Teaching Fellowship
- Student Research Grant [Awarded by Office of Diversity, Inclusion and Funding at **UW-Madison**]
- ISCA 2022 Student Travel Award
- Anita Borg Grace Hopper Celebration 2019 Scholarship
- Wisconsin Distinguished Fellow - Schneider
- Rajendra Prasad and Yasoda Mani Grandhi Fellow

Research Interests & Technical Skills

Research Interests : Computer Architecture, Microarchitecture, Memory Subsystem, Performance Modelling, Approximate Computing, Privacy-preserving Traces, Fully Homomorphic Encryption, Interconnects, Side-channel Attacks, AI Sustainability
Programming Languages : C++, Python, Bash, Verilog
Architectural Tools : ChampSim, PinTool, Perf, likwid, Gem5, Garnet, Sniper
Security Libraries : HELayers, OpenFHE

Relevant Coursework

Advanced Computer Architecture I and II, Introduction to Operating Systems, Introduction to Programming Languages and Compilers, Embedded Computing Systems, Neural Networks, Topics in Security and Privacy

Professional Service

Invited Talks

- Town Hall - University of Wisconsin - Madison, Dept. of Electrical and Computer Engineering, February 2025
- Computer Architecture Coordination Seminar Series - University of Central Florida, November 2023

Artifact Evaluation Committee

- IEEE/ACM International Symposium on Microarchitecture 2023 (MICRO 2023, 2022)
- Architectural Support for Programming Languages and Operating Systems (ASPLOS 2022)
- Conference on Machine Learning and Systems (MLSys 2020)
- Architectural Support for Programming Languages and Operating Systems (ASPLOS 2020)

Teaching

ECE 757 (Advanced Computer Architecture II)	Spring 2024, Spring 2025
ECE 352 (Digital System Fundamentals)	Fall 2024
ECE 5750 (High Performance Computer Architecture)	Fall 2017